

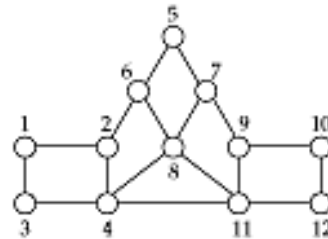
Routing in Telephone Networks

What is it?

- Process of finding a path from a source to every destination in the network
- Suppose you want to connect to Antarctica from your desktop
 - what route should you take?
 - does a shorter route exist?
 - what if a link along the route goes down?
 - what if you're on a mobile wireless link?
- Routing deals with these types of issues

Basics

- A *routing protocol* sets up a *routing table* in *routers* and *switch controllers*



ROUTING TABLE AT 1

Destination	Next hop	Destination	Next hop
1	—	7	2
2	2□	8□	2□
3	3□	9□	2□
4	3□	10□	2□
5	2□	11□	3□
6	2	12	3

- A node makes a *local* choice depending on *global* topology: this is the fundamental problem

Key problem

- How to make correct local decisions?
 - each router must know *something* about global state
- Global state
 - inherently large
 - dynamic
 - hard to collect
- *A routing protocol must intelligently summarize relevant information*

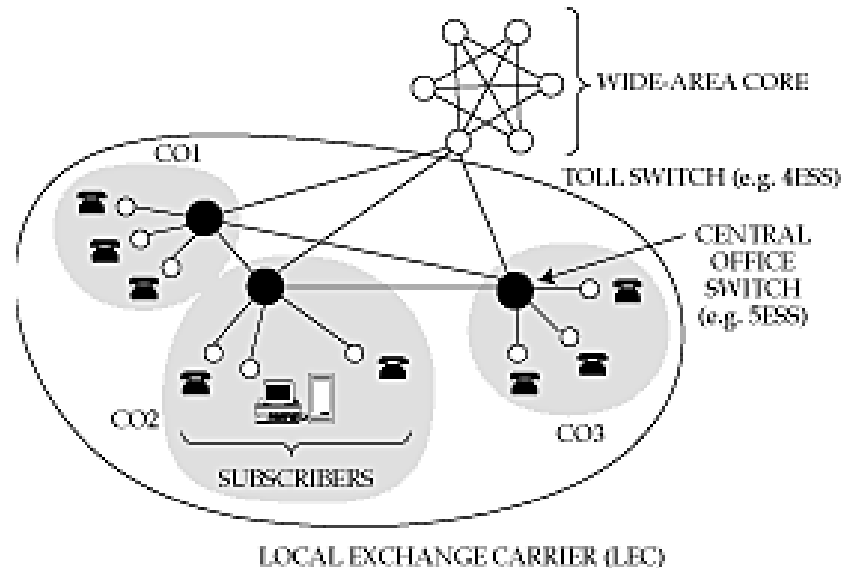
Requirements

- Minimize routing table space
 - fast to look up
 - less to exchange
- Minimize number and frequency of control messages
- Robustness: avoid
 - black holes
 - loops
 - oscillations
- Use optimal path

Choices

- Centralized vs. distributed routing
 - centralized is simpler, but prone to failure and congestion
- Source-based vs. hop-by-hop
 - how much is in packet header?
 - Intermediate: *loose source route*
- Stochastic vs. deterministic
 - stochastic spreads load, avoiding oscillations, but misorders
- Single vs. multiple path
 - primary and alternative paths (compare with stochastic)
- State-dependent vs. state-independent
 - do routes depend on current network state (e.g. delay)

Telephone network topology



- 3-level hierarchy, with a fully-connected core
- AT&T: 135 core switches with nearly 5 million circuits
- LECs may connect to multiple cores

Routing algorithm

- If endpoints are within same CO, directly connect
- If call is between COs in same LEC, use one-hop path between COs
- Otherwise send call to one of the cores
- Only major decision is at toll switch
 - one-hop or two-hop path to the destination toll switch
 - (why don't we need longer paths?)
- Essence of problem
 - which two-hop path to use if one-hop path is full

Features of telephone network routing

- Stable load
 - can predict pairwise load throughout the day
 - can choose optimal routes in advance
- Extremely reliable switches
 - downtime is less than a few minutes per year
 - can assume that a chosen route is available
 - can't do this in the Internet
- Single organization controls entire core
 - can collect global statistics and implement global changes
- Very highly connected network
- Connections require resources (but all need the same)

Statistics

- Poisson call arrival (independence assumption)
- Exponential call “holding” time (length!)
- Goal:- Minimise Call “Blocking” (aka “loss”) Probability subject to minimise cost of network

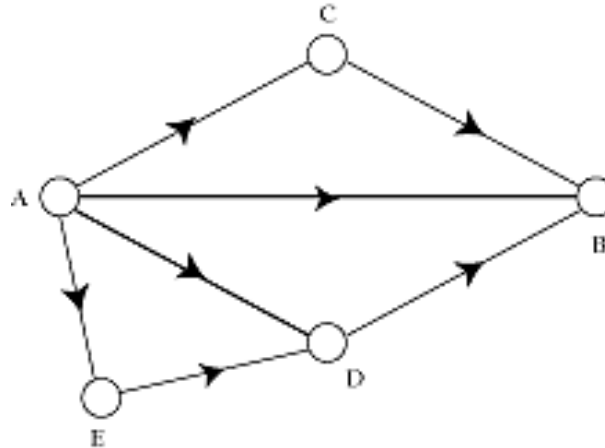
The cost of simplicity

- Simplicity of routing a historical necessity
- But requires
 - reliability in every component
 - logically fully-connected core
- Can we build an alternative that has same features as the telephone network, but is cheaper because it uses more sophisticated routing?
 - Yes: that is one of the motivations for ATM
 - But 80% of the cost is in the local loop
 - not affected by changes in core routing
 - Moreover, many of the software systems assume topology
 - too expensive to change them

Dynamic nonhierarchical routing (DNHR)

- Simplest core routing protocol
 - accept call if one-hop path is available, else drop
- DNHR
 - divides day into around 10-periods
 - in each period, each toll switch is assigned a primary one-hop path and a list of alternatives
 - can overflow to alternative if needed
 - drop only if all alternate paths are busy
 - *crankback*
- Problems
 - does not work well if actual traffic differs from prediction

Metastability



- Burst of activity can cause network to enter metastable state
 - high blocking probability even with a low load
- Removed by trunk reservation
 - prevents spilled traffic from taking over direct path

Trunk status map routing (TSMR)

- DNHR measures traffic once a week
- TSMR updates measurements once an hour or so
 - only if it changes “significantly”
- List of alternative paths is more up to date

Real-time network routing

- No centralized control
 - Each toll switch maintains a list of lightly loaded links
 - Intersection of source and destination lists gives set of lightly loaded paths
- Example
 - At A, list is C, D, E $\Rightarrow$ links AC, AD, AE lightly loaded
 - At B, list is D, F, G $\Rightarrow$ links BD, BF, BG lightly loaded
 - A asks B for its list
 - Intersection = D $\Rightarrow$ AD and BD lightly loaded $\Rightarrow$ ADB lightly loaded $\Rightarrow$ it is a good alternative path
- Very effective in practice: only about a couple of calls blocked in core out of about 250 million calls attempted every day

Dynamic Alternative Routing

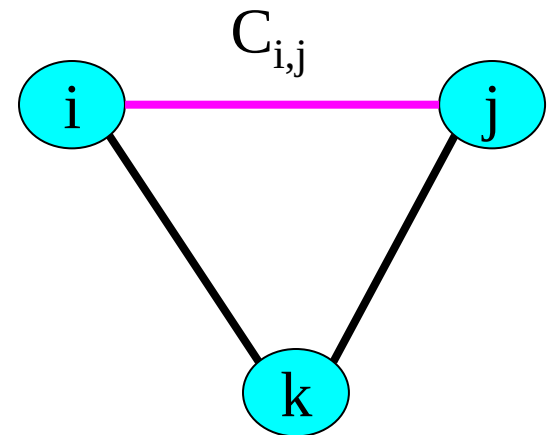
Very simple idea, but can be shown to provide optimal routes at very low complexity...

Underlying Network Properties

- Fully connected network
 - Underlying network is a trunk network
- Relatively small number of nodes
 - In 1986, the trunk network of British Telecom had only 50 nodes
 - Any algorithm with polynomial running time works fine
- Stochastic traffic
 - Low variance when the link is nearly saturated

Dynamic Alternative Routing

- Proposed by F.P. Kelly, R. Gibbens at British Telecom (well, Cambridge, Really:)
- Whenever the link (i, j) is saturated, use an alternative node (tandem)
- Q. How to choose tandem?



Fixed Tandem

- For any pair of nodes (i, j) we assign a fixed node k as tandem
- Needs careful traffic analysis and reprogramming
- Inflexible during breakdowns and unexpected traffic at tandem

Sticky Random Tandem

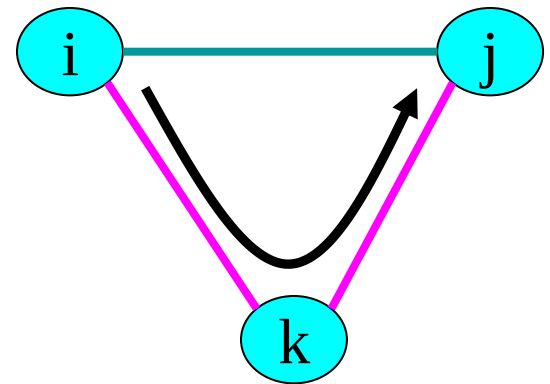
- If there is no free circuit along (i, j) , a new call is routed through a randomly chosen tandem k
- k is the tandem as long as it does not fail
- If k fails for a call, the call is lost and a new tandem is selected

Sticky Random Tandem

- Decentralized and flexible
 - No fancy pre-analysis of traffic required
 - Most of the time friendly tandems are used:
 - $p_k(i, j)$: proportion of calls between i and j which go through k
 - $q_k(i, j)$: proportion of calls that are blocked
-  $p_a(i, j)q_a(i, j) = p_b(i, j)q_b(i, j)$
- We may assign different frequencies to different tandems

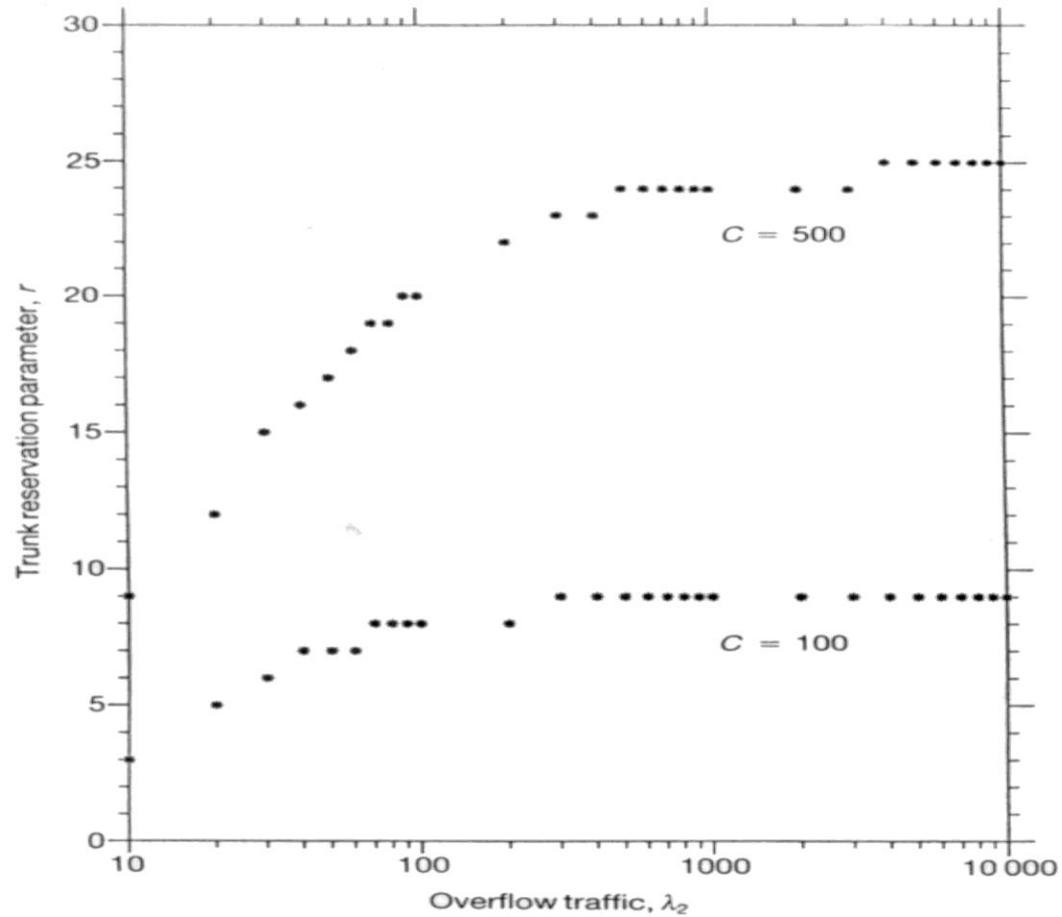
Trunk Reservation

- Unselfishness towards one's friends is good up to a point!!!
- We need to penalize two link calls, at least when the lines are very busy!



A tandem k accepts to forward calls if it has free capacity more than R

Trunk Reservation



Bounds: Erlang's Bound

- A node connected to C circuits
- Arrival: Poisson with mean v
- The expected value of blocking:

$$E(v, C) = \frac{v^C}{C!} \left[\sum_{i=0}^C \frac{v^i}{i!} \right]^{-1}$$

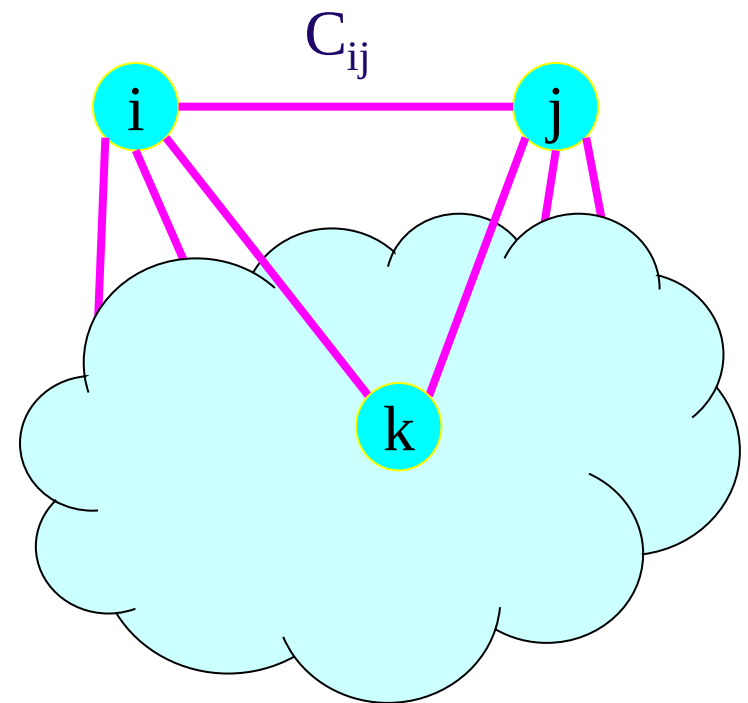
Max-flow Bound

- Capacity of (i, j) : C_{ij}
- Mean load on (i, j) :

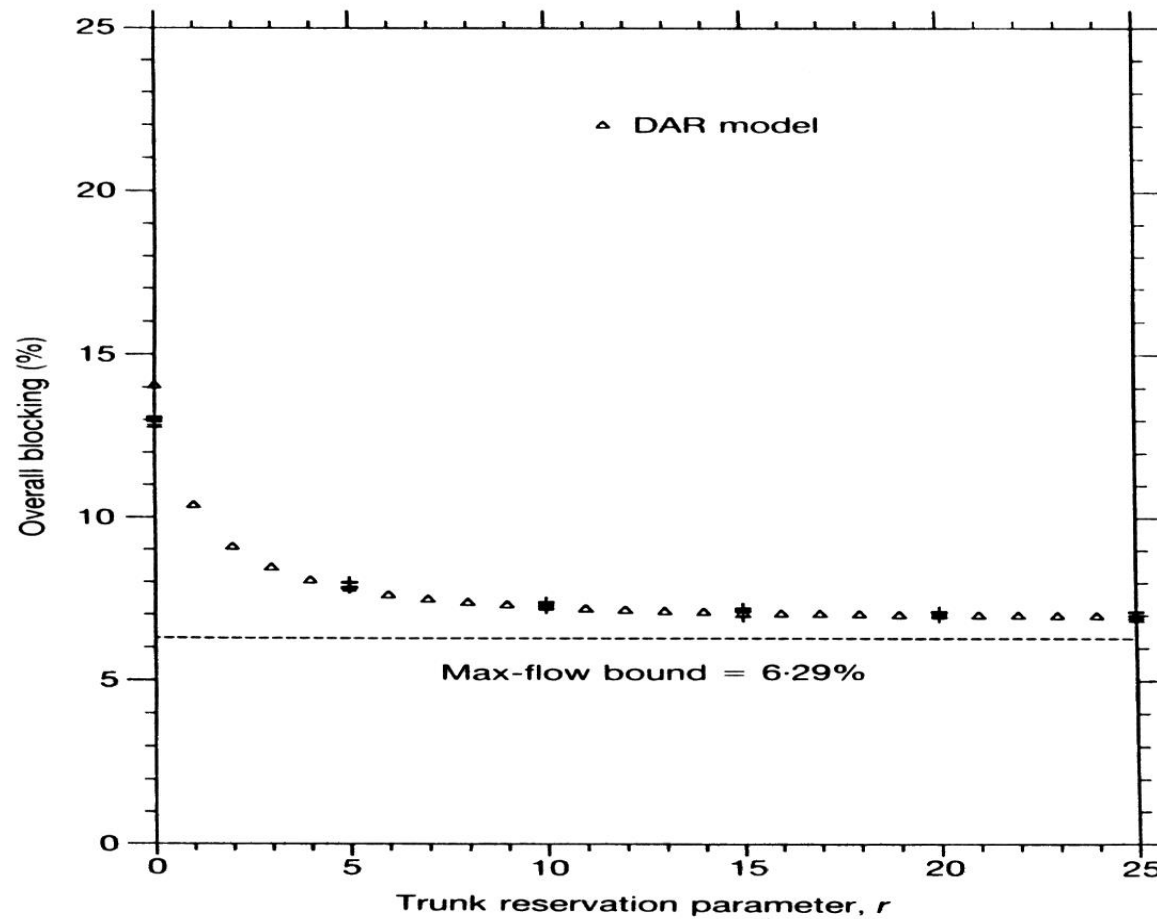
$$E \left[\sum_{i < j} n_{ij}(t) \right] \leq f(v(t))$$

- where f is:

$$\max \sum_{i < j} \left(x_{ij} + \sum_{k \neq i, j} x_{ikj} \right)$$

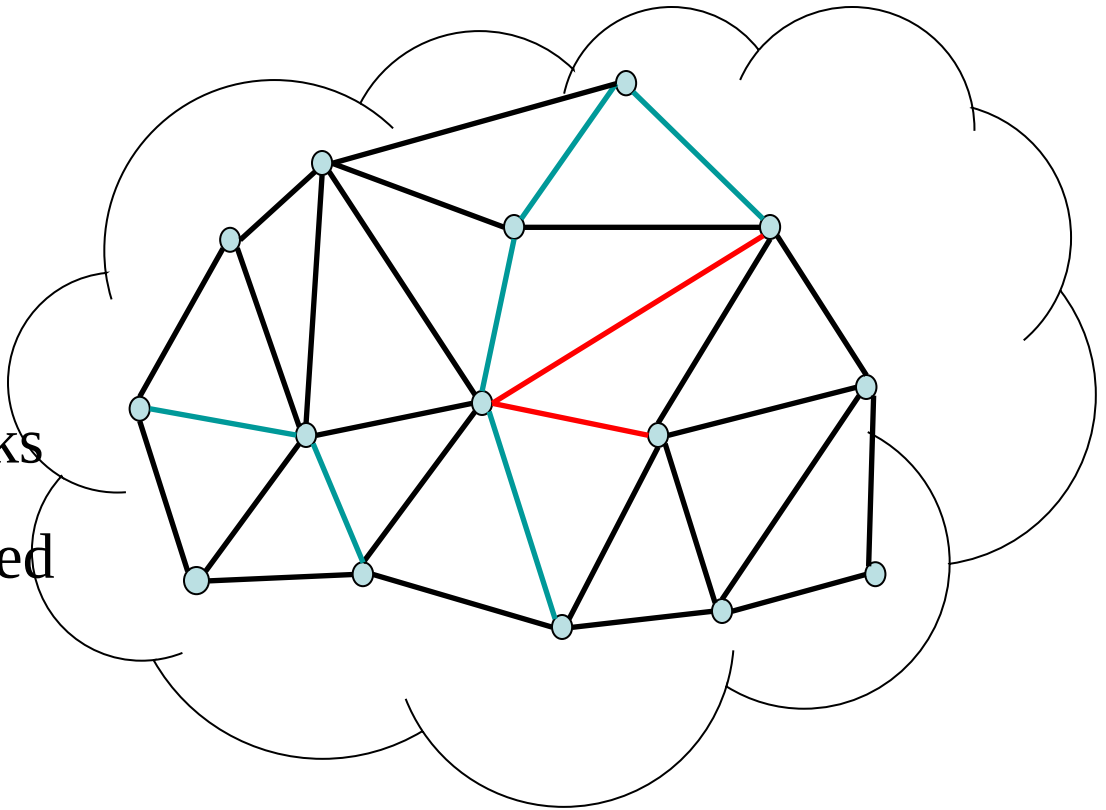


Trunk Reservation



Traffic, Capacity Mismatch

- Traffic > Capacity for some links
- Can we always find a feasible set of tandems?
- Red links: saturated links
- White links: not saturated
- Good triangle: one red, two white links



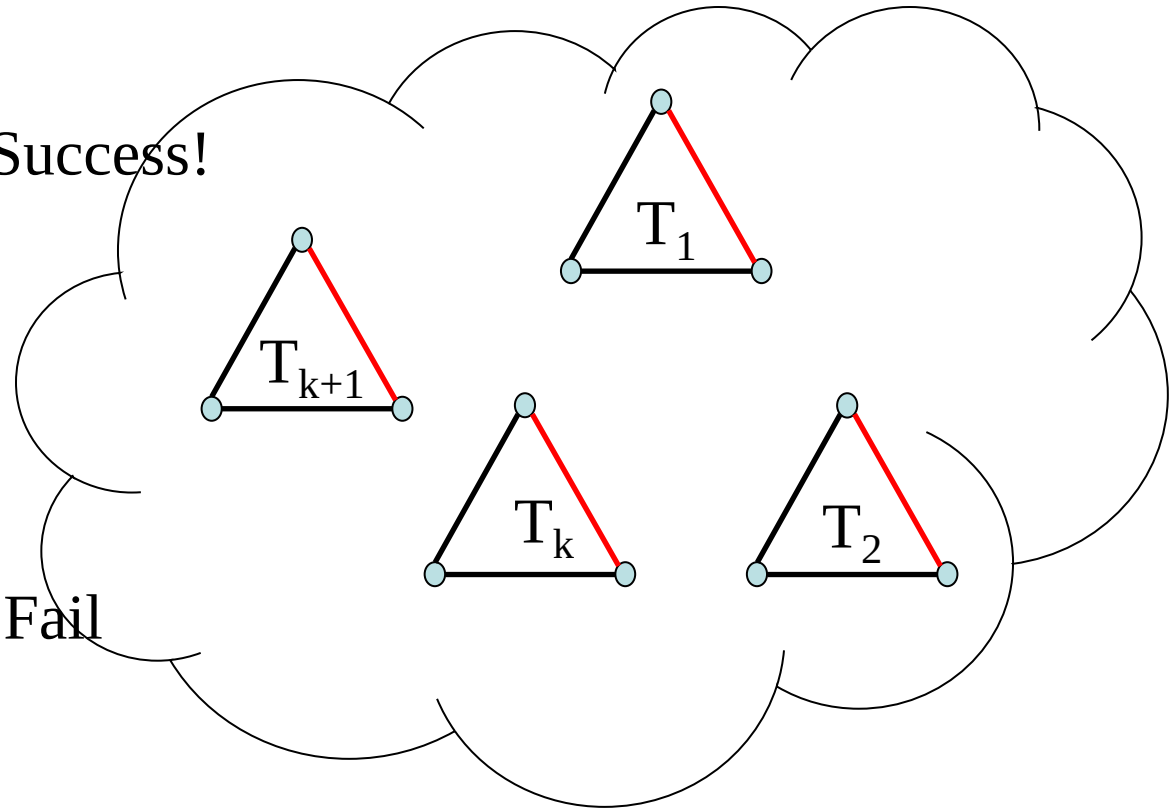
Greedy Algorithm

a. No *red links* $\Rightarrow$ Success!

b. Red link and a *good* triangle $\Rightarrow$ Success!

- Add good triangle to the list

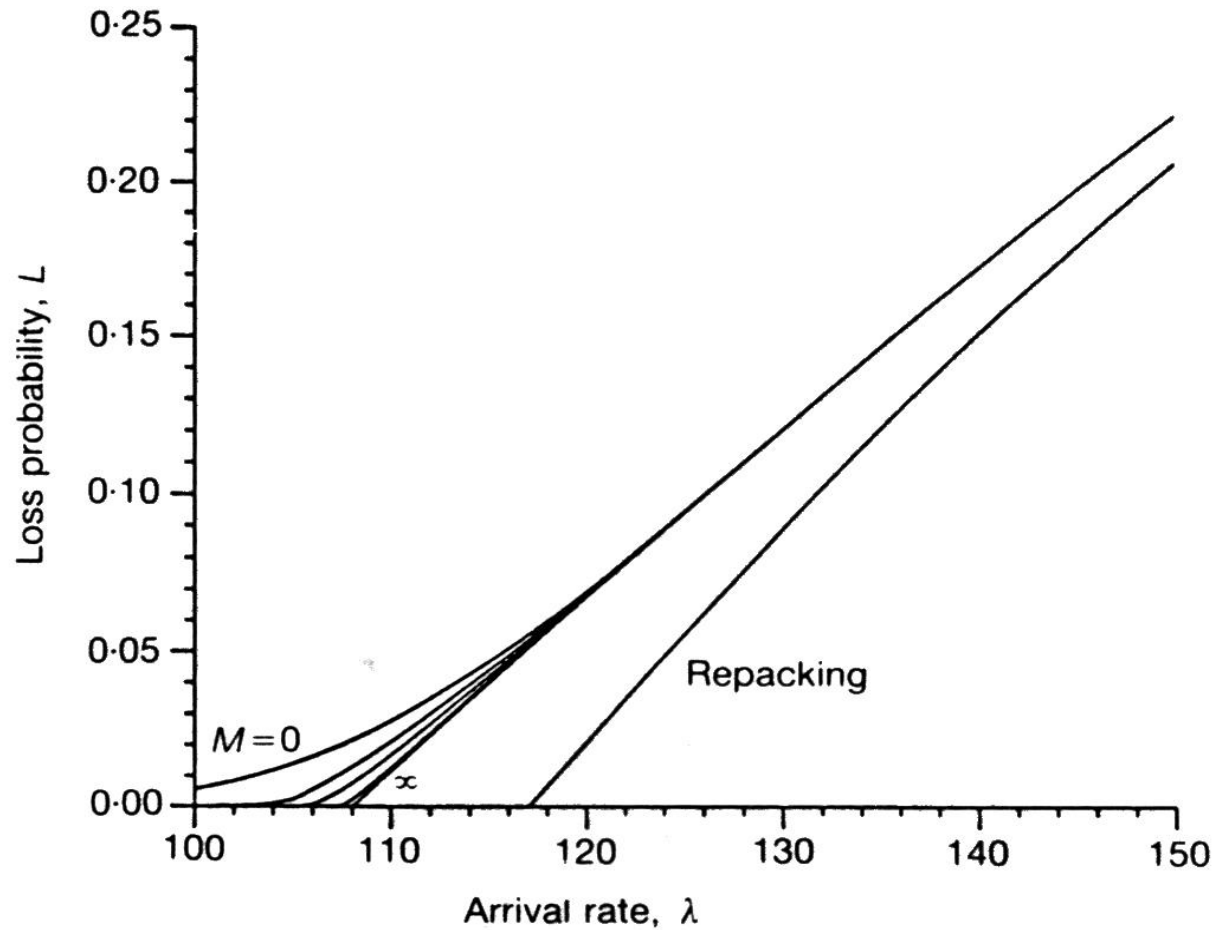
c. Red link and no good triangle $\Rightarrow$ Fail



Extensions to DAR

- *n*-link paths
 - Too much resources consumed, little benefit
- Multiple alternatives
 - *M* attempts before rejecting a call
- Least-busy alternative
- Repacking
 - A call in progress can be rerouted

Comparison of Extensions



Features of Internet Routing

- Packets, not circuits (
 - E.g. timescales can be much shorter
- Topology complicated/heterogeneous
- Many (10,000 ++) providers
- Traffic sources bursty
- Traffic matrix unpredictable
 - E.g. Not distance constrained
- Goal: maximise throughput, subject to min delay and cost (and energy?)

Internet Routing Model

- 2 key features:
 - Dynamic routing
 - Intra- and Inter-AS routing, AS = locus of admin control
- Internet organized as “*autonomous systems*” (AS).
 - AS is internally connected
- Interior Gateway Protocols (IGPs) within AS.
 - Eg: RIP, OSPF, HELLO
- Exterior Gateway Protocols (EGPs) for AS to AS routing.
 - Eg: EGP, BGP-4

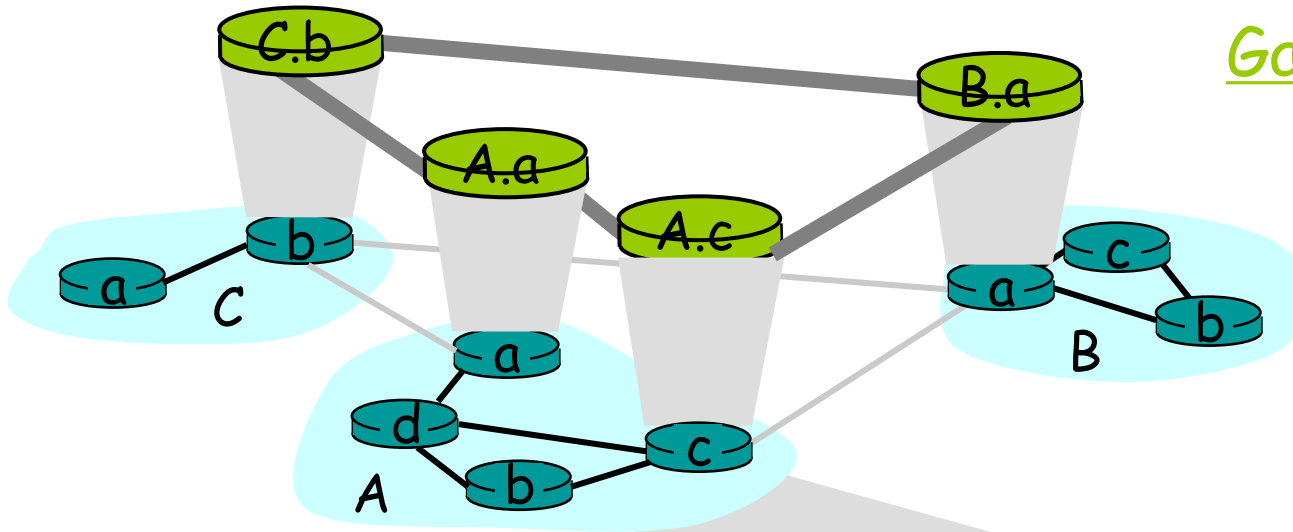
Requirements for Intra-AS Routing

- Should **scale** for the size of an AS.
 - Low end: 10s of routers (small enterprise)
 - High end: 1000s of routers (large ISP)
- Different requirements on **routing convergence** after topology changes
 - Low end: can tolerate some connectivity disruptions
 - High end: fast convergence essential to business (making money on transport)
- **Operational/Admin/Management (OAM) Complexity**
 - Low end: simple, self-configuring
 - High end: Self-configuring, but operator hooks for control
- **Traffic engineering** capabilities: high end only

Requirements for Inter-AS Routing

- Should scale for the size of the global Internet.
 - Focus on *reachability*, not optimality
 - Use *address aggregation* techniques to minimize core routing table sizes and associated control traffic
 - At the same time, it should allow *flexibility in topological structure* (eg: don't restrict to trees etc)
- Allow policy-based routing between autonomous systems
 - Policy refers to *arbitrary preference among a menu of available options* (based upon options' *attributes*)
 - In the case of routing, options include advertised AS-level routes to address prefixes
 - *Fully distributed routing* (as opposed to a signaled approach) is the only possibility.
 - *Extensible* to meet the demands for newer policies.

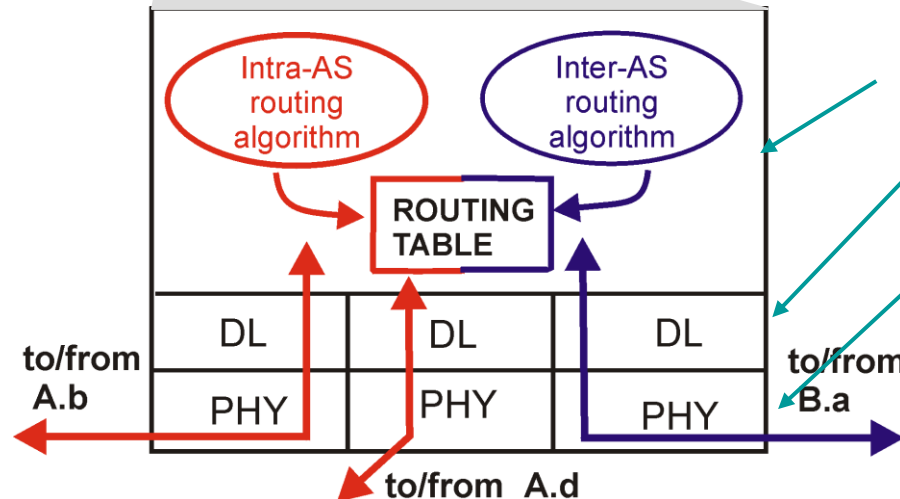
Intra-AS and Inter-AS routing



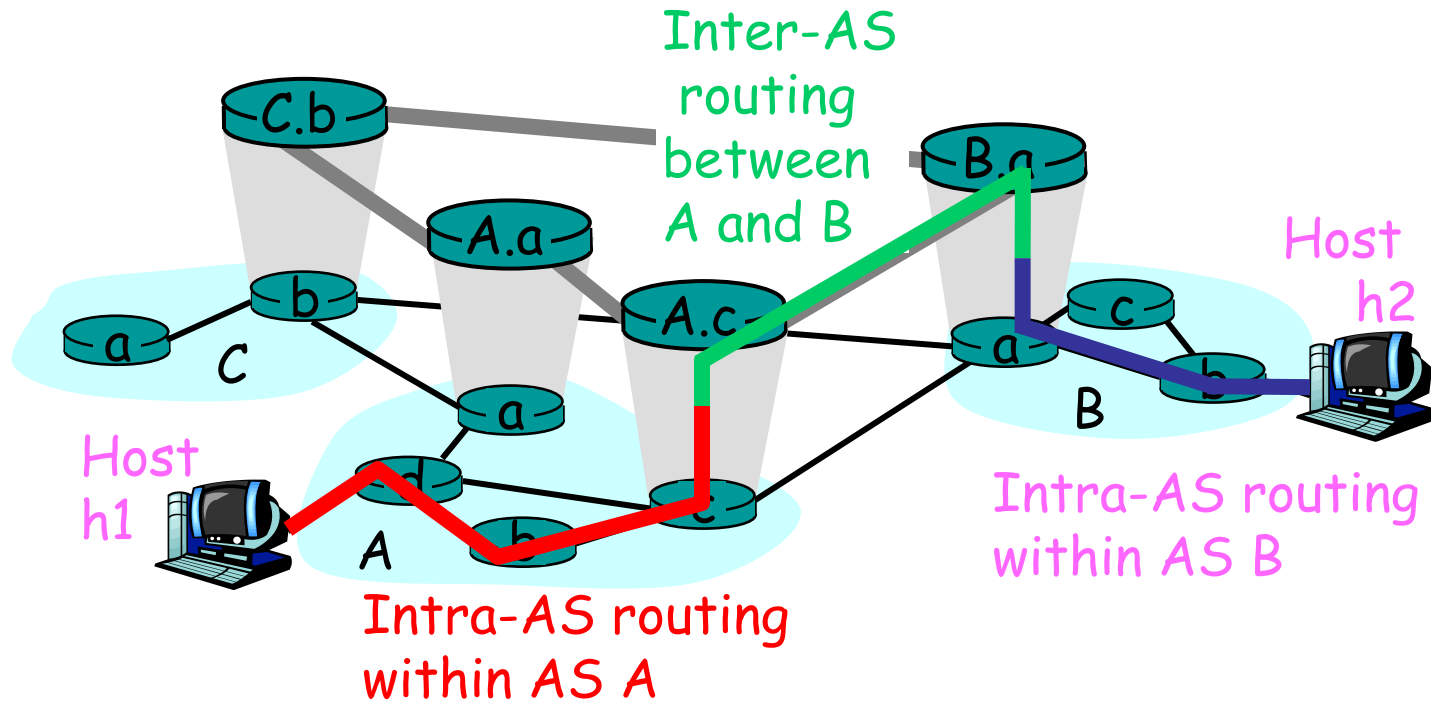
Gateways:

- perform **inter-AS** routing amongst themselves
- perform **intra-AS** routing with other routers in their AS

inter-AS,
intra-AS
routing in
gateway **A.c**



Intra-AS and Inter-AS routing: Example



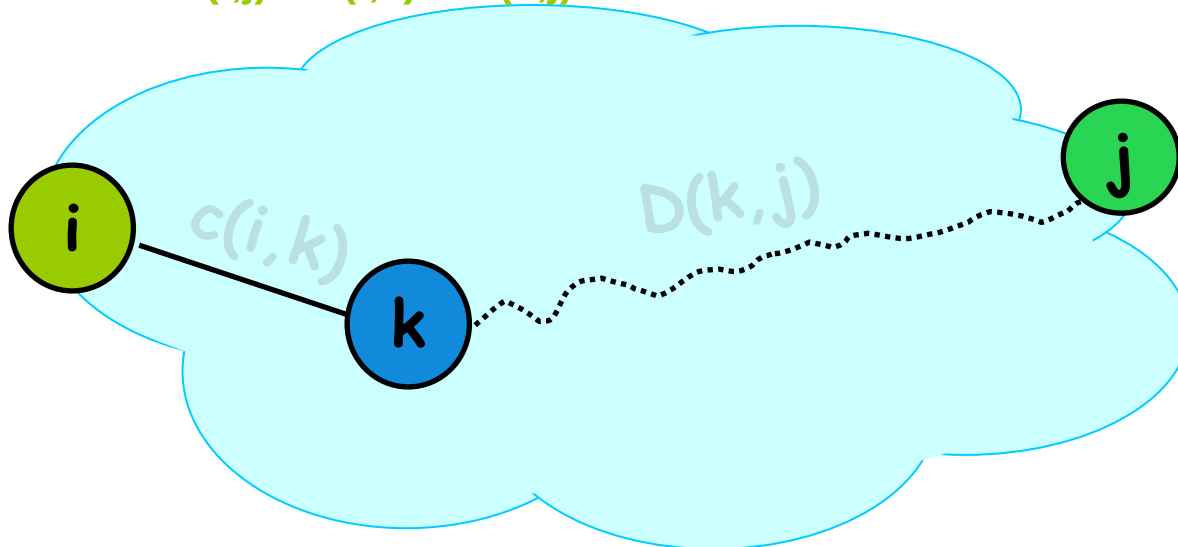
Basic Dynamic Routing Methods

- Source-based: source gets a map of the network,
 - source finds route, and either
 - signals the route-setup (eg: ATM approach)
 - encodes the route into packets (inefficient)
- Link state routing: per-link information
 - Get map of network (in terms of link states) at all nodes and find next-hops locally.
 - Maps consistent => next-hops consistent
- Distance vector: per-node information
 - At every node, set up distance signposts to destination nodes (a vector)
 - Setup this by peeking at neighbors' signposts.

DV & LS: consistency criterion

- *The subset of a shortest path is also the shortest path between the two intermediate nodes.*
- Corollary:
 - If the shortest path from node i to node j , with distance $D(i,j)$ passes through neighbor k , with link cost $c(i,k)$, then:

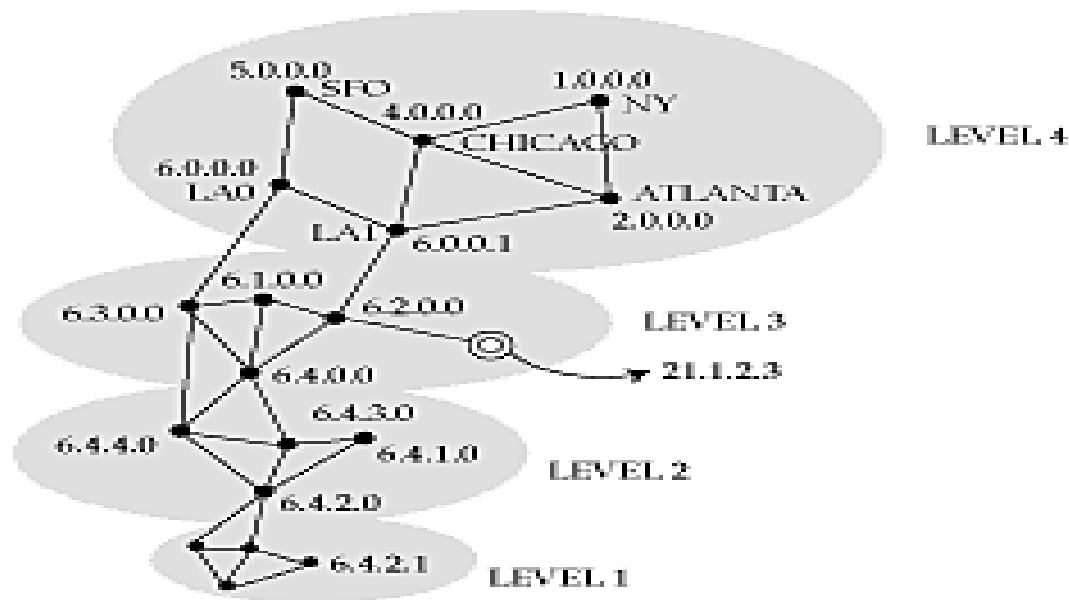
$$D(i,j) = c(i,k) + D(k,j)$$



Hierarchical routing

- Large networks need large routing tables
 - more computation to find shortest paths
 - more bandwidth wasted on exchanging DVs and LSPs
- Solution:
 - hierarchical routing
- Key idea
 - divide network into a set of domains
 - gateways connect domains
 - computers within domain unaware of outside computers
 - gateways know only about other gateways

Example



- Features
 - only a few routers in each level
 - not a strict hierarchy
 - gateways participate in multiple routing protocols
 - non-aggregable routers increase core table space

Hierarchy in the Internet

- Three-level hierarchy in addresses
 - network number
 - subnet number/more specific prefix
 - host number
- Core advertises routes only to networks, not to subnets
 - e.g. 135.104.*, 192.20.225.*
- Even so, about 80,000 networks in core routers (1996)
- Gateways talk to backbone to find best next-hop to every other network in the Internet

External and summary records

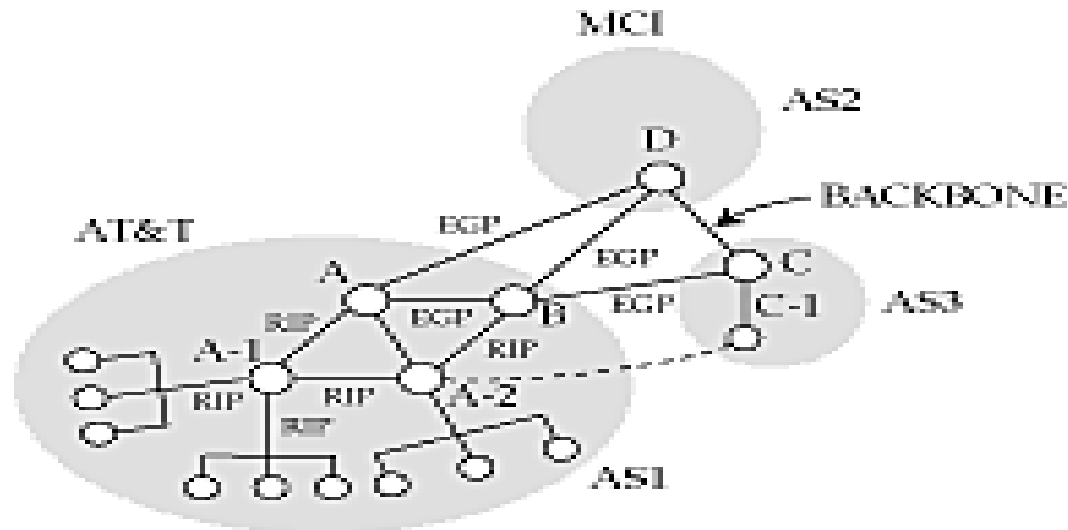
- If a domain has multiple gateways
 - *external* records tell hosts in a domain which one to pick to reach a host in an external domain
 - e.g allows 6.4.0.0 to discover shortest path to 5.* is through 6.0.0.0
 - *summary* records tell backbone which gateway to use to reach an internal node
 - e.g. allows 5.0.0.0 to discover shortest path to 6.4.0.0 is through 6.0.0.0
- External and summary records contain distance from gateway to external or internal node
 - unifies distance vector and link state algorithms

Interior and exterior protocols

- Internet has three levels of routing
 - highest is at *backbone* level, connecting *autonomous systems (AS)*
 - next level is within AS
 - lowest is within a LAN
- Protocol between AS gateways: exterior gateway protocol
- Protocol within AS: interior gateway protocol

Exterior gateway protocol

- Between untrusted routers
 - mutually suspicious
- Must tell a *border gateway* who can be trusted and what paths are allowed



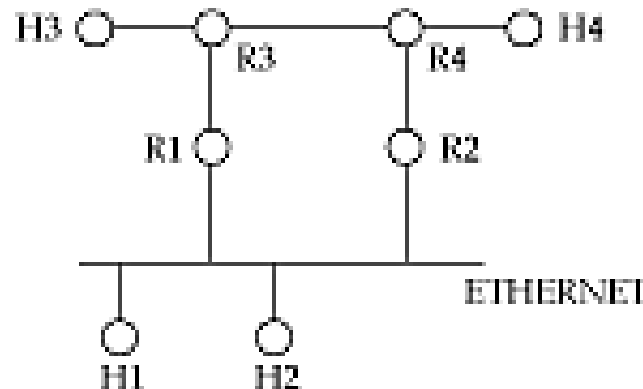
- *Transit over backdoors* is a problem

Interior protocols

- Much easier to implement
- Typically partition an AS into *areas*
- Exterior and summary records used between areas

Routing within a broadcast LAN

- What happens at an endpoint?
- On a point-to-point link, no problem
- On a broadcast LAN
 - is packet meant for destination within the LAN?
 - if so, what is the datalink address ?
 - if not, which router on the LAN to pick?
 - what is the router's datalink address?



Internet solution

- All hosts on the LAN have the same subnet address
- So, easy to determine if destination is on the same LAN
- Destination's datalink address determined using ARP
 - broadcast a request
 - owner of IP address replies
- To discover routers
 - routers periodically sends router advertisements
 - with preference level and time to live
 - pick most preferred router
 - delete overage records
 - can also force routers to reply with *solicitation message*

Redirection

- How to pick the best router?
- Send message to arbitrary router
- If that router's next hop is another router on the same LAN, host gets a *redirect* message
- It uses this for subsequent messages

Multicast routing

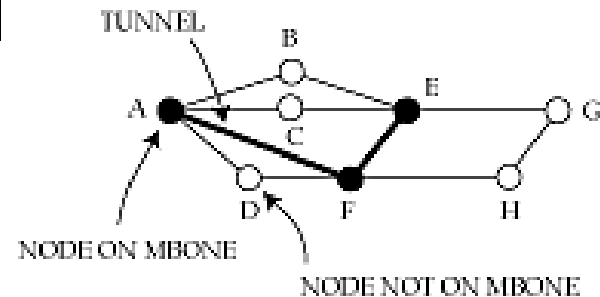
- Unicast: single source sends to a single destination
- Multicast: hosts are part of a *multicast group*
 - packet sent by *any* member of a group are received by *all*
- Useful for
 - multiparty videoconference
 - distance learning
 - resource location

Multicast flavors

- Unicast: point to point
- Multicast:
 - point to multipoint
 - multipoint to multipoint
- Can simulate point to multipoint by a set of point to point unicasts
- Can simulate multipoint to multipoint by a set of point to multipoint multicasts
- The difference is efficiency

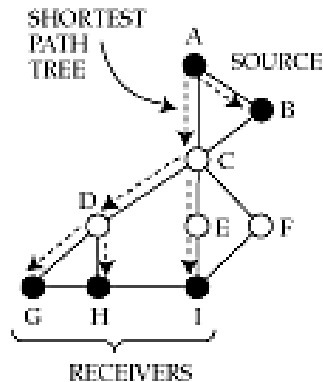
Multicast routing protocol

- Interface on “shortest path” to source depends on whether path is real or virtual



- Shortest path from E to A is not through C, but F
 - so packets from F will be flooded, but not from C
- Need to discover shortest paths only taking multicast-capable routers into account
 - Distance-vector Multicast routing protocol (DVMRP)

Shortest path tree



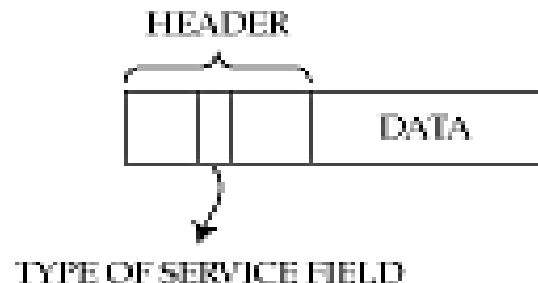
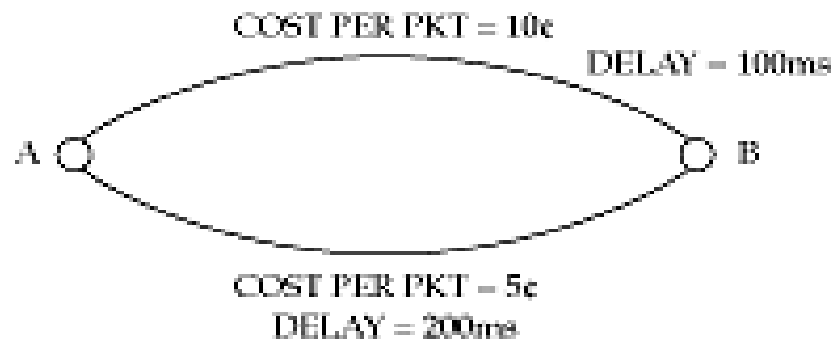
- Ideally, want to send exactly one multicast packet per link
 - forms a *multicast tree* rooted at sender
- Optimal multicast tree provides *shortest* path from sender to every receiver
 - *shortest-path* tree rooted at sender

Routing vs. policy routing

- In standard routing, a packet is forwarded on the ‘best’ path to destination
 - choice depends on load and link status
- With policy routing, routes are chosen depending on *policy* directives regarding things like
 - source and destination address
 - transit domains
 - quality of service
 - time of day
 - charging and accounting
- The general problem is still open
 - fine balance between correctness and information hiding

Multiple metrics

- Simplest approach to policy routing
- Advertise multiple costs per link
- Routers construct multiple shortest path trees



Problems with multiple metrics

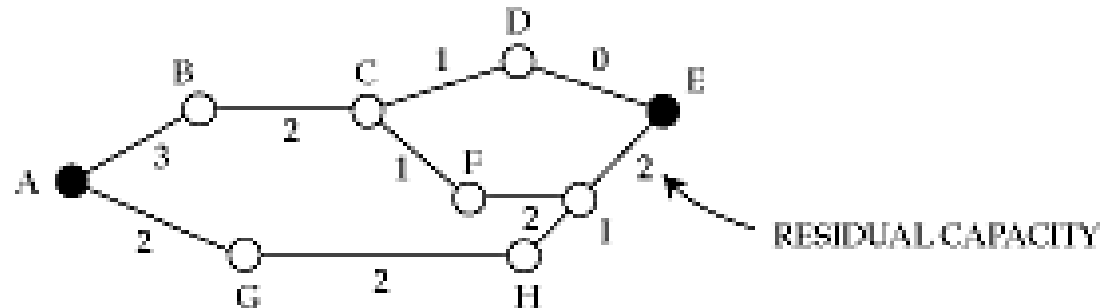
- All routers must use the same rule in computing paths
- Remote routers may misinterpret policy
 - source routing may solve this
 - but introduces other problems (what?)

Provider selection

- Another simple approach
- Assume that a single service provider provides almost all the path from source to destination
 - e.g. AT&T or MCI
- Then, choose policy simply by choosing provider
 - this could be dynamic (agents!)
- In Internet, can use a loose source route through service provider's access point
- Or, multiple addresses/names per host

Crankback

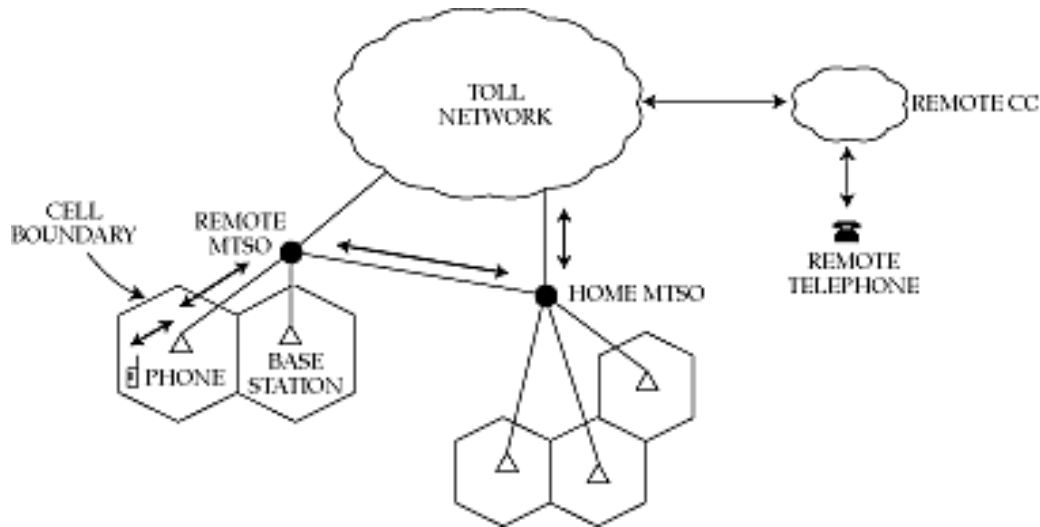
- Consider computing routes with QoS guarantees
- Router returns packet if no next hop with sufficient QoS can be found
- In ATM networks (PNNI) used for the call-setup packet
- In Internet, may need to be done for every packet!



Mobile routing

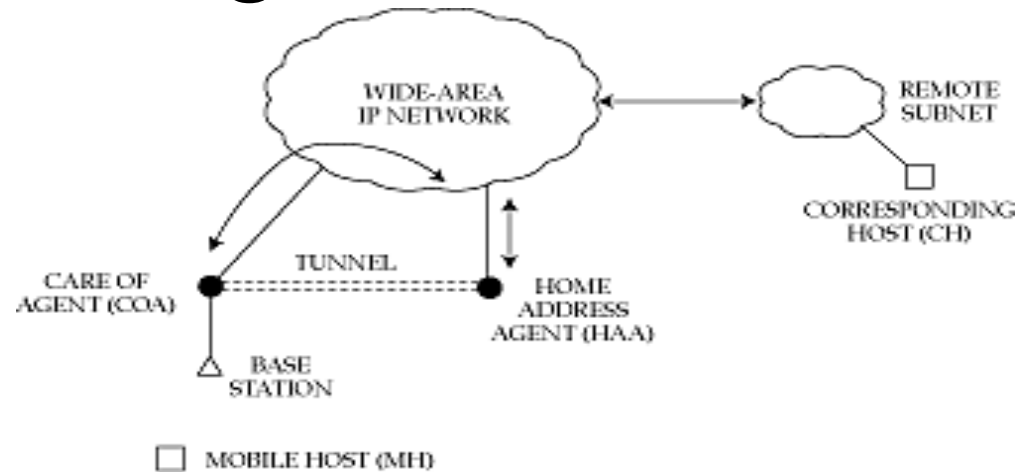
- How to find a mobile host?
- Two sub-problems
 - location (where is the host?)
 - routing (how to get packets to it?)
- We will study mobile routing in the Internet and in the telephone network

Mobile cellular routing



- Each cell phone has a global ID that it tells remote MTSO when turned on (using slotted ALOHA up channel)
- Remote MTSO tells home MTSO
- *To* phone: call forwarded to remote MTSO to closest base
- *From* phone: call forwarded to home MTSO from closest base
- New MTSOs can be added as load increases

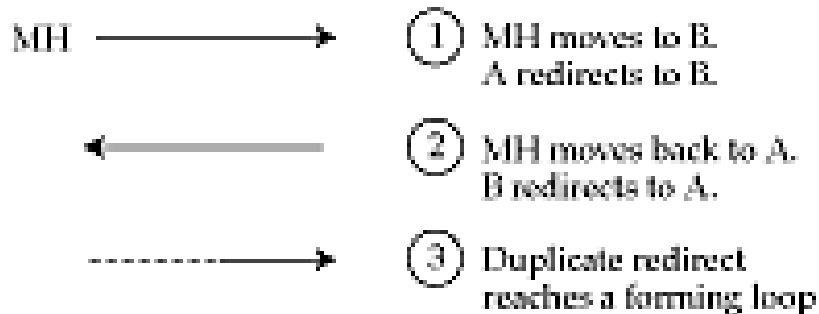
Mobile routing in the Internet



- Very similar to mobile telephony
 - but outgoing traffic does not go through home
 - and need to use tunnels to forward data
- Use *registration* packets instead of slotted ALOHA
 - passed on to home address agent
- Old care-of-agent forwards packets to new care-of-agent until home address agent learns of change

Problems

- Security
 - mobile and home address agent share a common secret
 - checked before forwarding packets to COA
- Loops



Thank You